

COLOMBO RATMALANA, Sri Lanka

WMO: 434670

Lat: 6.822N

Lon: 79.886E

Elev: 22

StdP: 14.68

Time Zone: 5.50 (IND)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	73.1	74.1	63.9	89.1	80.8	65.7	95.1	80.5	15.8	81.1	14.0	82.8	3.5	50	0.362

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	8.9	91.7	79.8	91.0	79.6	90.3	79.4	81.9	88.9	81.4	88.4	80.9	87.8	7.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.0	155.7	86.0	79.4	152.6	85.7	79.0	150.6	85.6	45.7	89.5	45.1	88.9	44.7	88.5	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
13.4	11.7	10.4	DB	67.3	96.7	2.5	4.1	65.4	99.6	64.0	102.0	62.6	104.3	60.7	107.3	
			WB	65.0	83.1	2.8	0.7	63.0	83.6	61.4	83.9	59.8	84.3	57.8	84.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	82.9	81.0	82.0	83.8	84.3	84.8	83.9	83.6	83.2	83.0	82.5	81.6	81.1	(6)
	DBStd	2.13	1.89	1.69	1.29	1.90	2.01	1.74	1.81	1.75	1.95	1.66	1.48	1.74	(7)
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
	CDD50	12011	962	897	1047	1030	1078	1017	1043	1028	989	1008	949	963	(10)
	CDD65	6536	497	477	582	580	613	567	578	563	539	543	499	498	(11)
	CDH74	75329	4976	5195	7195	7267	7899	7000	7009	6605	6219	6076	5026	4862	(12)
	CDH80	28855	1658	1893	3117	3314	3580	2789	2676	2310	2184	2178	1639	1517	(13)
(14) Wind	WSAvg	4.7	5.4	4.6	3.8	3.5	5.1	5.6	5.6	5.6	4.9	3.9	3.7	4.8	(14)
(15) Precipitation	PrecAvg	97.2	2.9	3.0	4.8	9.6	14.1	7.6	5.4	4.1	8.9	14.5	13.2	6.7	(15)
	PrecMax	154.3	8.3	8.7	14.1	24.3	29.6	23.7	19.0	10.7	21.7	34.3	38.2	18.8	(16)
	PrecMin	69.8	0.0	0.0	0.0	2.8	1.9	1.2	0.4	0.1	0.8	3.7	2.3	0.7	(17)
	PrecStd	18.9	2.2	2.4	2.9	5.2	7.3	4.0	4.2	2.8	5.2	6.6	6.5	4.3	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	90.9	91.7	92.3	92.9	92.0	90.0	89.9	89.3	89.8	89.9	89.9	90.4	(19)
		MCWB	74.9	76.8	78.7	81.3	81.3	80.5	80.1	79.7	79.5	78.8	77.3	75.4	(20)
	2%	DB	89.4	89.9	91.3	91.8	91.0	89.1	88.7	88.2	88.9	89.1	88.9	88.7	(21)
		MCWB	75.4	77.2	78.9	80.8	81.0	80.3	79.5	79.2	79.2	78.8	77.6	76.1	(22)
	5%	DB	88.1	88.8	90.5	91.1	90.3	88.4	87.8	87.4	88.0	88.2	88.1	87.6	(23)
		MCWB	75.7	77.2	78.8	80.5	80.8	80.1	79.2	78.9	78.9	78.6	77.8	76.6	(24)
	10%	DB	86.9	87.7	89.7	90.3	89.4	87.6	87.0	86.6	87.1	87.5	87.2	86.6	(25)
		MCWB	75.6	77.1	78.7	80.3	80.5	79.8	79.0	78.7	78.7	78.4	77.8	76.8	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.6	80.1	81.0	82.6	82.8	81.8	80.9	80.7	80.7	80.5	80.0	79.7	(27)
		MCDB	86.3	87.0	88.8	90.6	89.3	87.5	88.6	88.1	87.6	86.5	86.3	85.5	(28)
	2%	WB	78.3	79.4	80.5	81.8	81.9	81.0	80.1	80.0	80.0	79.9	79.2	79.0	(29)
		MCDB	85.3	86.4	88.7	89.8	88.5	87.0	86.7	86.5	86.5	86.0	85.7	85.0	(30)
	5%	WB	77.8	78.8	80.0	81.3	81.6	80.7	79.8	79.5	79.5	79.3	78.8	78.4	(31)
		MCDB	84.6	85.9	88.0	89.1	88.1	86.7	86.2	85.7	85.8	85.6	85.5	84.3	(32)
	10%	WB	77.1	78.2	79.5	80.8	81.0	80.2	79.3	79.1	79.1	79.0	78.3	77.9	(33)
		MCDB	83.8	85.2	87.4	88.5	87.5	86.1	85.5	85.1	85.1	85.4	84.6	83.7	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	12.7	12.4	12.4	12.0	8.9	7.3	6.8	7.1	8.0	10.0	11.2	11.2	(35)
		MCDBR	13.8	13.3	13.0	12.6	9.9	8.1	7.3	8.1	9.1	10.7	12.4	12.7	(36)
		MCWBR	5.5	5.0	4.4	4.7	3.6	2.9	2.6	2.8	2.9	3.5	4.1	4.6	(37)
	5% WB	MCDBR	11.9	12.3	12.3	12.0	9.3	7.6	7.3	7.8	8.5	10.1	11.7	11.5	(38)
		MCWBR	5.2	5.2	4.6	5.0	3.8	3.1	3.0	3.0	3.3	3.8	4.2	4.7	(39)
(40) Clear-Sky Solar Irradiance	taub		0.438	0.454	0.497	0.500	0.516	0.498	0.514	0.494	0.463	0.470	0.469	0.455	(40)
	taud		2.364	2.302	2.196	2.214	2.157	2.222	2.150	2.226	2.330	2.306	2.304	2.330	(41)
	Ebn at Noon		279	277	265	260	250	252	249	259	270	269	267	271	(42)
	Edh at Noon		40	43	49	47	48	44	48	46	42	43	42	41	(43)
(44) All-Sky Solar Radiation	RadAvg		1690	1895	1993	1881	1684	1676	1717	1749	1764	1687	1497	1504	(44)
	RadStd		118	102	93	96	126	99	81	89	122	106	107	127	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	+0.31	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page